

Experiment 20: Future Sales Prediction

Aim

To implement **Future Sales Prediction** using a suitable machine learning algorithm.

Algorithm

Step 1: Import the required libraries.

Step 2: Create or load the sales dataset.

Step 3: Prepare the input feature and target sales values.

Step 4: Split the dataset into training and testing sets.

Step 5: Create a Linear Regression model.

Step 6: Train the model using the training data.

Step 7: Predict future sales values.

Step 8: Calculate the R^2 score.

Step 9: Display the predicted sales and model performance.

Program

```
#Experiment 20: Future Sales Prediction
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

data = pd.read_csv("sales.csv")

X = data[["Month"]]
y = data["Sales"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Predicted Sales:", y_pred)
print("R2 Score:", r2_score(y_test, y_pred))

future_month = [[13]]
future_sales = model.predict(future_month)

print("Predicted Sales for Month 13:", future_sales[0])
```

Output

```
... Predicted Sales: [268.45670391 253.19832402 115.87290503]
R2 Score: 0.9977655312351841
Predicted Sales for Month 13: 298.9734636871508
```

Result

Thus, the **Future Sales Prediction model** was successfully implemented using **Linear Regression**, and future sales were predicted based on the historical sales data.